

SaaS

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Official WhatsApp Business API + Chatbot + Inbox + CRM + Campaigns

Project Handout — From Zero to Launch Plan

یہ handout project folder میں رکھنے کے لیے بنایا گیا ہے۔ اس میں business direction, phases, architecture, database, security, deployment, beta launch اور proof checklist شامل ہے۔

Important Decision: API key seller نہیں بننا؛ **official WhatsApp automation** **SaaS provider** بننا ہے۔

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Executive Summary .1

اس project کا مقصد ایک professional SaaS platform بنانا ہے جہاں businesses اپنا official WhatsApp Business account/number connect کر کے customer support, chatbot automation, CRM, templates اور campaigns manage کر سکیں۔

- Product کو raw API key selling کے طور پر launch نہیں کرنا۔
- client کا اپنا Meta Business / WABA / WhatsApp number ہونا چاہیے۔
- آپ dashboard, automation, inbox, integrations, reporting اور support sell کریں گے۔
- Compliance, opt-in, security, encrypted tokens اور tenant isolation لازمی ہوں گے۔

Correct Business Positioning .2

غلط طریقہ	صحیح طریقہ
Raw API key یا shared token بیچنا	Official WhatsApp Automation SaaS sell کرنا
ایک number/token کو multiple clients کو دینا	ہر client کا اپنا WABA/phone number connect کروانا
Spam sender یا bulk blast tool بنانا	Opt-in based campaign + CRM + support tool بنانا
Unofficial QR/WhatsApp Web automation	Official Meta WhatsApp Business Platform use کرنا

Professional market wording

- Official WhatsApp Business API Setup + Automation Dashboard
- WhatsApp Chatbot, CRM Inbox & Business Automation SaaS
- Connect your official WhatsApp Cloud API and automate support, leads, orders, and campaigns

Phase-wise Roadmap .3

Main Output	Focus	Phase
SaaS provider model، API key seller نہیں	Business direction lock	Phase 0
Target customers, pricing, beta clients	Market validation	Phase 1

Developer app, test number, webhook test	Official Meta setup	Phase 2
Next.js, PostgreSQL, Prisma, Auth, hosting plan	Tech stack	Phase 3
Multi-tenant schema + secure token storage	Database design	Phase 4
Landing, dashboard, inbox, settings, admin	MVP screens	Phase 5
WABA ID, phone ID, token, test send	Manual WhatsApp connection	Phase 6
Incoming messages saved and shown	Webhook system	Phase 7
WhatsApp reply سے Dashboard	Send message system	Phase 8
Live chat inbox	Inbox MVP	Phase 9
Keyword/reply automation	Basic chatbot	Phase 10
Approved template management	Templates	Phase 11
Opt-in based campaigns	Contacts + campaigns	Phase 12
Plans, limits, usage logs	Billing + usage	Phase 13
Encryption, isolation, rate limits	Security hardening	Phase 14
Live production setup	Deployment	Phase 15
real clients 5-3	Beta launch	Phase 16
Automated onboarding	Embedded Signup	Phase 17
Reseller/agency model	Agency white-label	Phase 18

Phase 0 — Business Direction Lock .4

Goal: project کی **official** اور **safe direction lock** کرنا.

.Offer: Official WhatsApp API setup + Chatbot + Inbox + CRM + Campaigns •

Avoid: raw API keys, rented numbers, shared tokens, unofficial WhatsApp Web •
.automation

Target customers: ecommerce stores, clinics/salons, real estate, service businesses, •
.agencies

.Primary revenue: setup fee + monthly SaaS subscription + add-ons •

Phase 1 — Market Validation .5

Goal: coding سے پہلے **demand verify** کرنا۔

1. 20 potential clients سے بات کریں۔

2. ہر client سے پوچھیں کہ روز WhatsApp پر کتنے messages آتے ہیں۔

3. Auto reply, team inbox, campaign, order notification اور CRM needs note کریں۔

4. 3 beta clients identify کریں جو real testing کر سکیں۔

5. Starter, Growth, Agency pricing estimate بنائیں۔

.Output: target customer, beta clients, pricing, must-have features

Phase 2 — Official Meta Setup .6

Goal: Meta Developer app اور **WhatsApp test environment ready** کرنا۔

• Meta Business Portfolio

• Meta Developer App

• WhatsApp product

• Test WABA and test phone number

• Webhook verify endpoint

• Send/receive test message

:SMS verification issue handling

• Fresh business number use کریں۔

• SMS اور voice call دونوں receive ہونے چاہئیں۔

• Number پہلے WhatsApp/Cloud API میں conflict نہ کر رہا ہو۔

• SMS fail ہو تو voice call verification try کریں۔

• Client کو checklist دیں: SIM active, SMS blocking off, call forwarding off, number accessible

Phase 3 — Recommended Tech Stack .7

Recommended Option	Area
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Next.js + Tailwind CSS	Frontend
Node.js/NestJS یا Next.js API Routes	Backend
PostgreSQL	Database
Prisma	ORM
Supabase Auth یا NextAuth/Auth.js, Clerk	Auth
local manual invoice initially یا Stripe/Paddle	Payments
Vercel + Railway/Render/Supabase	Hosting
BullMQ + Redis / Upstash	Queue later
Sentry / Logtail	Logs later

MVP میں simple monorepo رکھیں۔ پہلے complexity کم رکھیں، پھر scale کریں۔

Phase 4 — Database Design .8

Goal: multi-client SaaS structure بنانا جہاں ہر **client** کا **data isolated** ہو۔

- users •
- tenants •
- team_members •
- whatsapp_accounts •
- phone_numbers •
- access_tokens •
- contacts •
- conversations •
- messages •
- message_templates •
- automations •
- campaigns •
- webhook_events •
- usage_logs •
- invoices •

audit_logs •

Critical rule: **business record** میں **tenantId** لازمی ہو۔ ایک **client** کا **data** دوسرے **client** کو کبھی نظر نہیں آنا چاہیے۔

Phase 5 — MVP Screens .9

Admin Dashboard	Client Dashboard	Public Website
All clients	Dashboard overview	Home page
Connected numbers	Connect WhatsApp	Pricing page
Message usage	Inbox	Features page
Failed webhooks	Contacts	Contact page
Subscription status	Templates	Login/signup
Abuse/spam monitoring	Settings + Billing	Privacy + Terms

Phase 6 — Manual WhatsApp Connection MVP .10

شروع میں Embedded Signup کے بجائے manual setup رکھیں تاکہ API flow مکمل سمجھ آ جائے۔

Client Business Name •

WABA ID •

Phone Number ID •

Access Token — encrypted storage only •

Webhook Verify Token •

Connection Status •

Send Test Message button •

Incoming Webhook Test •

Important: token کبھی **frontend** میں **show** نہ ہو۔ صرف **connection status** دکھائیں۔

Core Product Build — Webhook, Messages, Inbox, Bot, .11

Templates, Campaigns

Webhook System 11.1

- api/webhooks/whatsapp endpoint/
- GET request سے webhook verification
- POST payload کو webhook_events میں save کرنا
- Incoming message extract کر کے save contact/conversation/message کرنا
- Processing status اور error logs save کرنا

Send Message System 11.2

- Backend tenant ownership verify کرے
- WhatsApp API call backend سے ہو
- Message status save ہو: sent, delivered, read, failed
- Error user-friendly انداز میں dashboard پر show ہو

Inbox MVP 11.3

- Conversation list
- Customer profile
- Message thread
- Reply box
- Unread count
- Agent assignment later

Basic Chatbot Builder 11.4

- Keyword trigger
- Welcome message
- Fallback message

Office-hours reply •

Human handoff •

Templates + Campaigns 11.5

Approved template list •

Variables preview •

Contacts import •

Opt-in status required •

Unsubscribe/STOP handling •

Campaign failure report •

Phase 13 — Billing + Usage .12

Included	Best For	Plan
number, basic inbox, basic bot, 1 limited contacts	Small shops	Starter
Team inbox, automations, templates, campaigns	Growing businesses	Growth
Multiple clients, white-label, reports	Marketing agencies	Agency

.Usage logs: messages sent, delivered, failed, category, tenantId, billing period •

Setup fee • ضرور رکھیں کیونکہ onboarding اور verification میں time لگتا ہے۔

Phase 14 — Security Hardening .13

Access tokens encrypted storage •

Tokens frontend میں کبھی نہ جائیں •

Tenant-based data isolation •

Auth-protected dashboard •

Backend-level admin route protection •

Webhook verification •

Rate limiting •

Audit logs •

CORS production domains only •

Secrets logs/error messages میں نہ آئیں •

env. کبھی GitHub پر upload نہ ہو •

Database backup plan •

Phase 15 — Deployment Plan .14

:Recommended deployment

Frontend: Vercel •

Backend/API: Vercel functions, Railway یا Render •

Database: Supabase/Railway PostgreSQL •

Domain + SSL: Cloudflare/Vercel •

Logs: Sentry •

Email: Resend/SendGrid •

:Deployment checklist

Production .env set •

Database migrations applied •

Webhook URL live domain پر set •

HTTPS working •

CORS production domain allowed •

Build pass •

Login working •

Dashboard working •

Send/receive message working •

Logs visible •

Phase 16 — Beta Launch .15

- صرف 3-5 clients onboard کریں۔
- client کا اپنا WhatsApp number ہو۔
- Contacts کے opt-in کی written confirmation لیں۔
- Daily logs check کریں۔
- Failed messages investigate کریں۔
- Client feedback collect کریں۔
- Bugs fix کریں۔
- Pricing validate کریں۔

Beta success criteria: 7 دن webhook stable رہے، messages send/receive ہوں، tenant data leak نہ ہو، اور کم از کم 1 client paid plan پر آئے۔

Phase 17–18 — Embedded Signup + Agency White-label .16

- MVP stable ہونے کے بعد manual connection کو Embedded Signup میں upgrade کریں۔ پھر agency/white-label model add کریں۔
- Connect WhatsApp button
 - Meta login flow
 - Business/WABA/phone assets capture
 - Token exchange
 - Auto database setup
 - Custom agency logo/domain
 - Agency clients, limits, billing, reports

Day Execution Timeline-90 .17

Expected Result	Main Focus	Time
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Beta clients + send/receive test	Market validation + Meta test setup	Days 1–15
MVP foundation	Auth, database, dashboard, manual connect	Days 16–30
Live chat usable	Webhook, send message, inbox	Days 31–45
Automation layer	Bot, templates, contacts	Days 46–60
Paid SaaS structure	Campaigns, billing, admin	Days 61–75
Production beta	Deployment, security, beta launch	Days 76–90

Phase 18 – Proof Gate – قبول کرنے سے پہلے

کسی **phase** کو **complete/done/ready** تب تک قبول نہ کریں جب تک **evidence** نہ ملے۔

- Files changed •
- Commands run •
- Pass/fail output •
- Build/lint/type-check/test result •
- API endpoint proof if backend involved •
- Browser page proof if frontend involved •
- Console error check •
- Screenshot proof for UI •
- Database/migration proof if DB involved •
- Remaining issues, risks, next recommended step •

Main Risks and Controls .19

Control	Risk
Fresh number, SMS/call readiness checklist, voice call option	SMS verification fail
Official API, opt-in, no spam, compliance logs	Meta account ban
Encryption, no frontend exposure, restricted admin access	Token leak

tenantId everywhere, backend authorization checks	Tenant data leakage
Usage logs, limits, billing alerts	High message cost
Raw event logs, retry status, monitoring	Webhook instability
Campaign limits, abuse detection, opt-out enforcement	Client misuse

Final Project Manager Decision .20

.Approved direction: Official WhatsApp Automation SaaS Provider

Rejected direction: Raw API key seller, shared token seller, unofficial QR automation, or spam sender

Coding کریں۔ Phase 2 Meta test setup complete اور Next practical step: Phase 1 market validation
تب شروع کریں جب official test API سے send/receive message successfully ہو جائے۔

Official References .21

/WhatsApp Business Platform: <https://whatsappbusiness.com/products/business-platform> •

/WhatsApp Business Messaging Policy: <https://business.whatsapp.com/policy> •

/Meta WhatsApp Cloud API Docs: <https://developers.facebook.com/docs/whatsapp> •

WhatsApp Webhooks: •

<https://developers.facebook.com/docs/whatsapp/cloud-api/webhooks>

/Embedded Signup: <https://developers.facebook.com/docs/whatsapp/embedded-signup> •

/WhatsApp Platform Pricing: <https://whatsappbusiness.com/products/platform-pricing> •